

Anomaly Detection and Application to Intrusion Detection

Introduction

- ◆ We are drowning in the deluge of data that are being collected world-wide, while starving for knowledge at the same time*
- ◆ Anomalous events occur relatively infrequently
- ◆ However, when they do occur, their consequences can be quite dramatic and quite often in a negative sense

What are Anomalies?

- Anomaly is a pattern in the data that does not conform to the expected behavior
- Also referred to as outliers, exceptions, peculiarities, surprises, etc.
- Anomalies translate to significant (often critical) real life entities
 - Cyber intrusions
 - Credit card fraud
 - Faults in mechanical systems

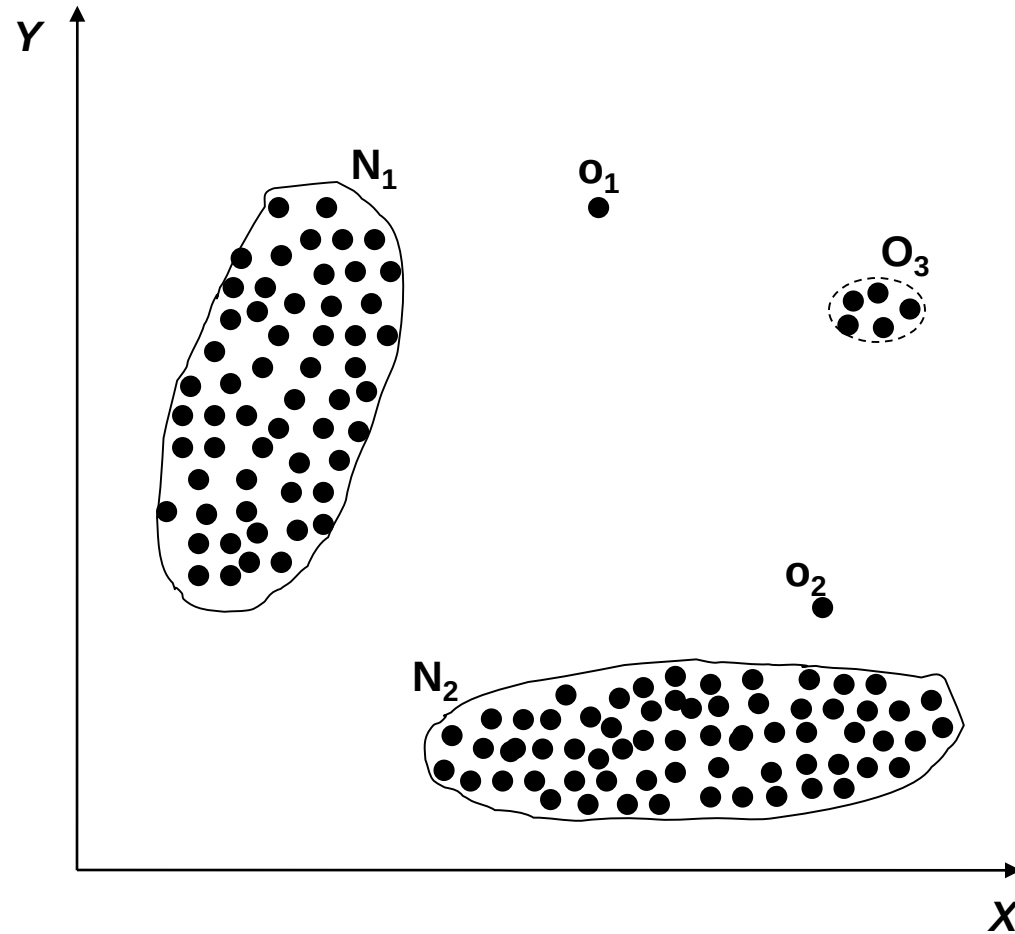
Real World Anomalies

- Credit Card Fraud
 - An abnormally high purchase made on a credit card
- Cyber Intrusions
 - A web server involved in *ftp* traffic



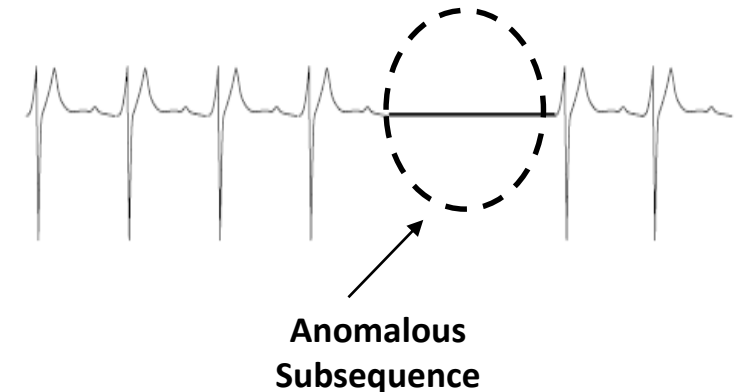
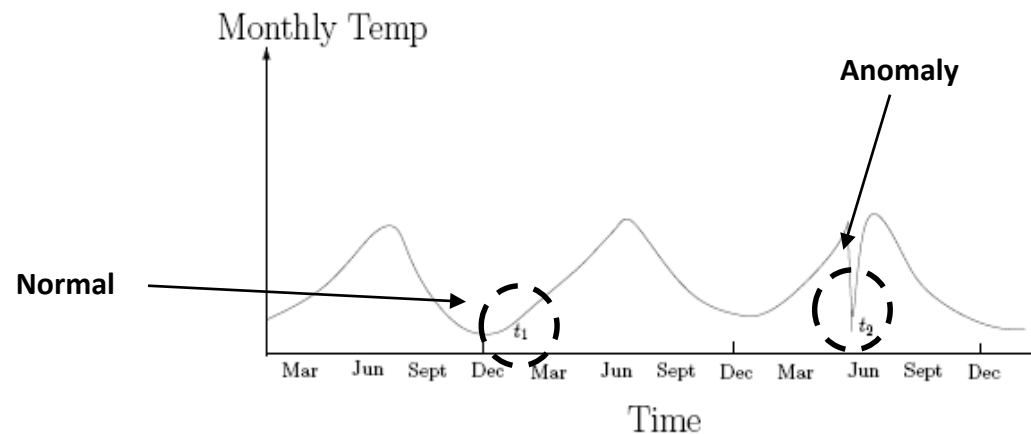
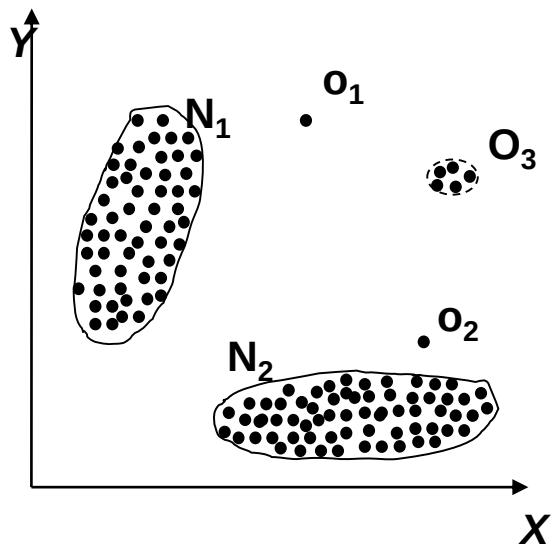
Simple Examples

- N_1 and N_2 are regions of normal behavior
- Points o_1 and o_2 are anomalies
- Points in region O_3 are also anomalies



Type of Anomalies

- **Point Anomalies:** An individual data instance is anomalous w.r.t. the data
- **Contextual Anomalies:** An individual data instance is anomalous within a context. Requires a notion of context Also referred to as conditional anomalies
- **Collective Anomalies:** A collection of related data instances is anomalous The individual instances within a collective anomaly are not anomalous by themselves. Requires a relationship among data instances
 - Sequential Data
 - Spatial Data
 - Graph Data



Key Challenges of Anomaly Detection

- **Defining a representative normal region** is challenging
- The **boundary between normal and outlying** behavior is often not precise
- **Availability of labeled data** for training/validation
- The **exact notion of an outlier** is different for different application domains
- Data might contain **noise**
- Normal behavior keeps evolving
- Appropriate **selection of relevant features**

Input Data

- Most common form of data handled by anomaly detection techniques is

Record Data

- Univariate
- Multivariate

Engine Temperature
192
195
180
199
19
177
172
285
195
163

Univariate

Multivariate

<i>Tid</i>	SrcIP	Start time	Dest IP	Dest Port	Number of bytes	Attack
1	206.135.38.95	11:07:20	160.94.179.223	139	192	No
2	206.163.37.95	11:13:56	160.94.179.219	139	195	No
3	206.163.37.95	11:14:29	160.94.179.217	139	180	No
4	206.163.37.95	11:14:30	160.94.179.255	139	199	No
5	206.163.37.95	11:14:32	160.94.179.254	139	19	Yes
6	206.163.37.95	11:14:35	160.94.179.253	139	177	No
7	206.163.37.95	11:14:36	160.94.179.252	139	172	No
8	206.163.37.95	11:14:38	160.94.179.251	139	285	Yes
9	206.163.37.95	11:14:41	160.94.179.250	139	195	No
10	206.163.37.95	11:14:44	160.94.179.249	139	163	Yes

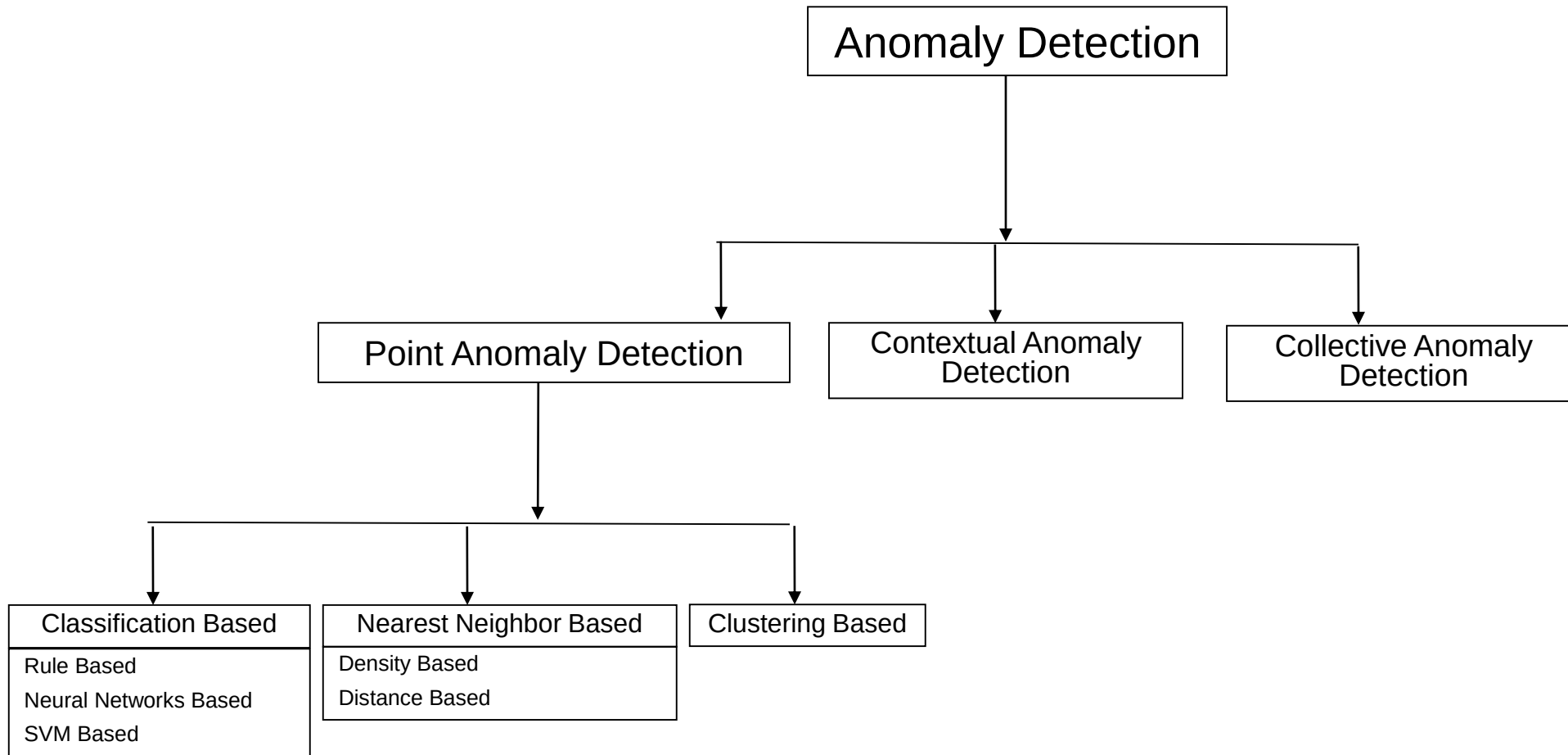
Data Labels

- Supervised Anomaly Detection
 - Labels available for both normal data and anomalies
 - Similar to rare class mining
 - Machine learning models: Naïve Bayes, Neural Network
- Semi-supervised Anomaly Detection
 - Labels available only for normal data
- Unsupervised Anomaly Detection
 - No labels assumed
 - Based on the assumption that anomalies are very rare compared to normal data
 - Machine learning: clustering

Output of Anomaly Detection

- Label
 - Each test instance is given a *normal* or *anomaly* label
 - This is especially true of classification-based approaches
- Score
 - Each test instance is assigned an anomaly score
 - Allows the output to be ranked
 - Requires an additional threshold parameter

Taxonomy



Classification Based Techniques

- Main idea: build a classification model for normal (and anomalous (rare)) events based on labeled training data, and use it to classify each new unseen event
- Classification models must be able to handle skewed (imbalanced) class distributions
- Categories:
 - *Supervised classification techniques*
 - Require knowledge of both **normal** and **anomaly** class
 - Build classifier to distinguish between normal and known anomalies

Nearest Neighbor Based Techniques

- *Key assumption*: normal points have close neighbors while anomalies are located far from other points
- General two-step approach
 1. Compute neighborhood for each data record
 2. Analyze the neighborhood to determine whether data record is anomaly or not
- Categories:
 - Distance based methods
 - Anomalies are data points most distant from other points
 - Density based methods
 - Anomalies are data points in low density regions

Nearest Neighbor Based Techniques

- Advantage

- Can be used in unsupervised or semi-supervised setting (do not make any assumptions about data distribution)

- Drawbacks

- If normal points do not have sufficient number of neighbors the techniques may fail
- Computationally expensive
- In high dimensional spaces, data is sparse and the concept of similarity may not be meaningful anymore. Due to the sparseness, distances between any two data records may become quite similar => Each data record may be considered as potential outlier!

Nearest Neighbor Based Techniques

- Distance based approaches
 - A point O in a dataset is an $DB(p, d)$ outlier if at least fraction p of the points in the data set lies greater than distance d from the point O^*
- Density based approaches
 - Compute local densities of particular regions and declare instances in low density regions as potential anomalies
 - Approaches
 - Local Outlier Factor (LOF)
 - Connectivity Outlier Factor (COF)
 - Multi-Granularity Deviation Factor (MDEF)

Evaluation of Anomaly Detection – F-value

- ♦ Accuracy is not sufficient metric for evaluation
 - Example: network traffic data set with 99.9% of normal data and 0.1% of intrusions
 - Trivial classifier that labels everything with the normal class can achieve 99.9% accuracy !!!!!

Confusion matrix		Predicted class	
		NC	C
Actual class	NC	TN	FP
	C	FN	TP

anomaly class – C
normal class – NC

- **Focus on both recall and precision**
 - Recall (R) = $TP / (TP + FN)$
 - Precision (P) = $TP / (TP + FP)$
- **F – measure = $2 * R * P / (R + P)$**

Applications of Anomaly Detection

- Network intrusion detection
- Insurance / Credit card fraud detection
- Healthcare Informatics / Medical diagnostics
- Industrial Damage Detection
- Image Processing / Video surveillance
- Novel Topic Detection in Text Mining

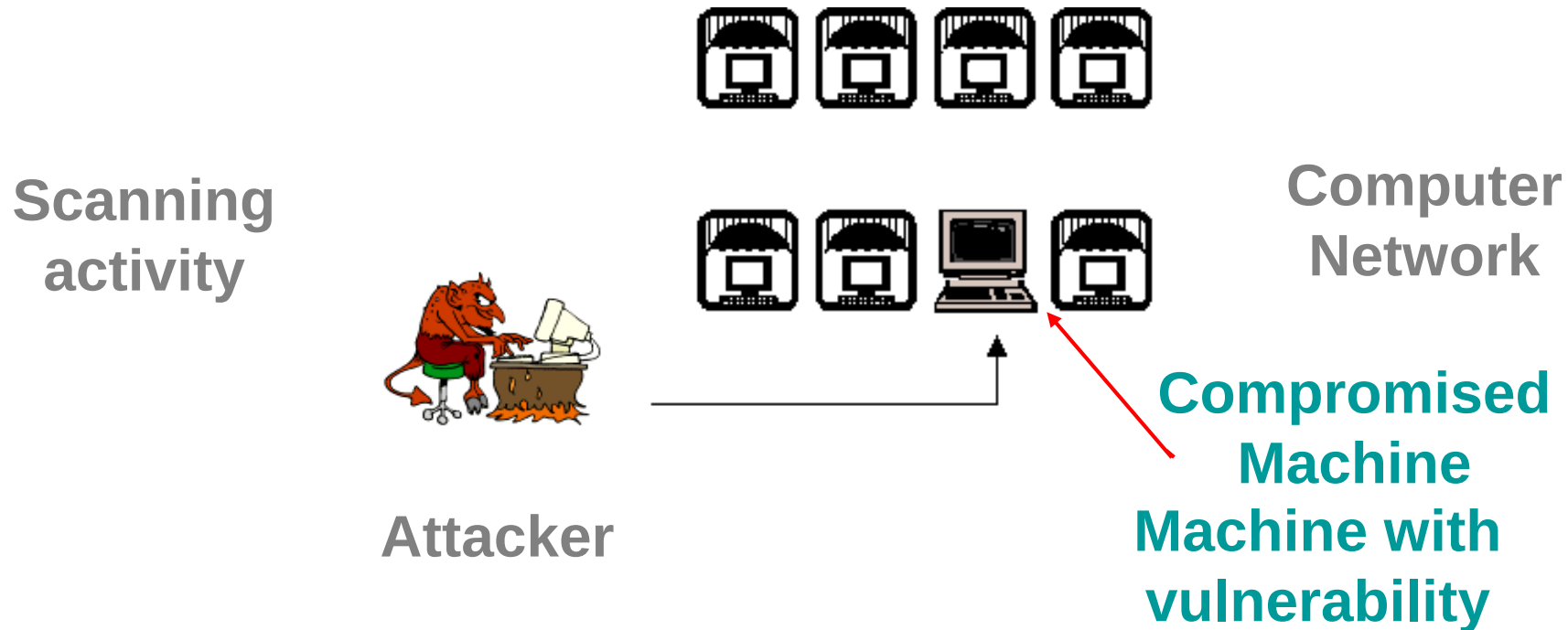
Intrusion Detection

- Intrusion Detection:
 - Process of monitoring the events occurring in a computer system or network and analyzing them for intrusions
 - Intrusions are defined as attempts to bypass the security mechanisms of a computer or network
- Challenges
 - Traditional signature-based intrusion detection systems are based on signatures of known attacks and cannot detect emerging cyber threats
 - Substantial latency in deployment of newly created signatures across the computer system
- Anomaly detection can alleviate these limitations



What are Intrusions?

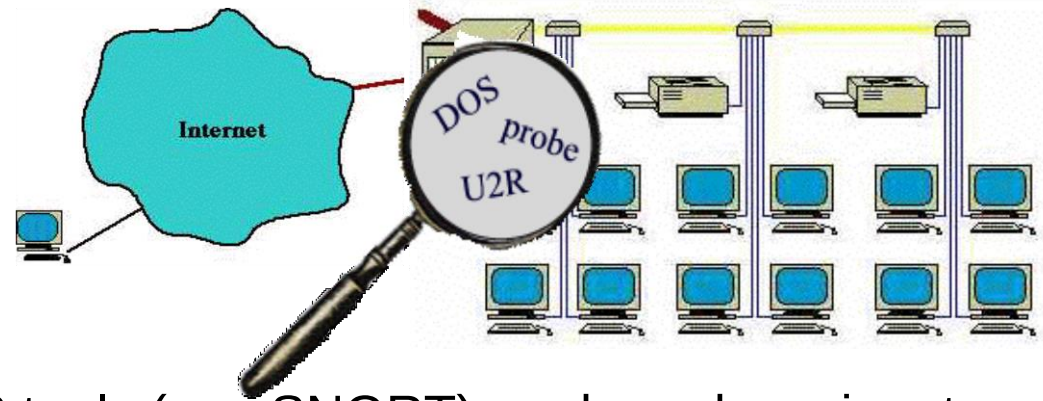
- ◆ Intrusions are actions that attempt to bypass security mechanisms of computer systems. They are usually caused by:
 - Attackers accessing the system from Internet
 - Insider attackers - authorized users attempting to gain and misuse non-authorized privileges
- ◆ Typical intrusion scenario



Intrusion Detection

♦ Intrusion Detection System

- combination of software and hardware that attempts to perform intrusion detection
- raises the alarm when possible intrusion happens



♦ Traditional intrusion detection system IDS tools (e.g. SNORT) are based on signatures of known attacks

- Example of SNORT rule (MS-SQL “Slammer” worm)

```
any -> udp port 1434 (content:"|81 F1 03 01 04 9B 81 F1 01|";  
content:"sock"; content:"send")
```



www.snort.org

♦ Limitations

- Signature database has to be manually revised for each new type of discovered intrusion
- They cannot detect emerging cyber threats
- Substantial latency in deployment of newly created signatures across the computer system

Approaches for Intrusion Detection

- ◆ Increased interest in machine learning based intrusion detection
 - Attacks for which it is difficult to build signatures
 - Attack stealthiness
 - Unforeseen/Unknown/Emerging attacks
 - Distributed/coordinated attacks
- ◆ Approaches for intrusion detection
 - *Misuse detection (Classification)*
 - ◆ Building predictive models from labeled data sets (instances are labeled as “normal” or “intrusive”) to identify known intrusions
 - ◆ High accuracy in detecting many kinds of known attacks
 - ◆ Cannot detect unknown and emerging attacks
 - *Apply Anomaly detection*
 - ◆ Detect novel attacks as deviations from “normal” behavior
 - ◆ Potential high false alarm rate - previously unseen (yet legitimate) system behaviors may also be recognized as anomalies
 - *Summarization of network traffic (Rule-based)*

Approaches for Intrusion Detection

Misuse Detection –
Building Predictive
Models

Tid	SrcIP	Start time	Dest IP	Dest Port	Number of bytes	Attack
1	206.135.38.95	11:07:20	160.94.179.223	139	192	No
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3	206.163.37.95	11:14:29	160.94.179.217	139	180	No
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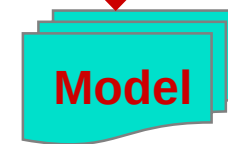
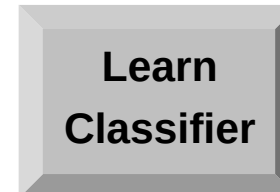
*Summarization of
attacks using
association rules*

Rules Discovered:

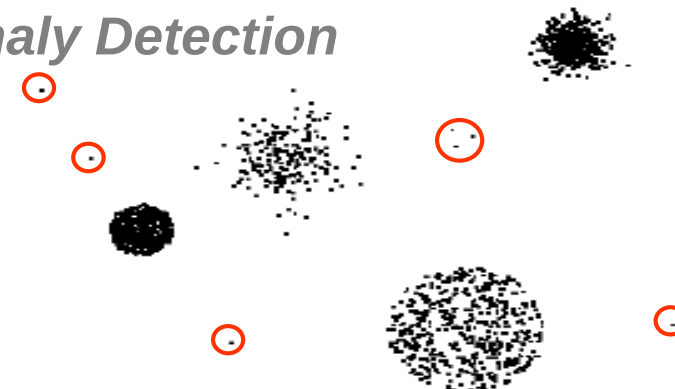
**{Src IP = 206.163.37.95,
Dest Port = 139,
Bytes ∈ [150, 200]} --> {ATTACK}**

categorical
temporal
categorical
continuous
class

Tid	SrcIP	Start time	Dest IP	Number of bytes	Attack
1	206.163.37.81	11:17:51	160.94.179.208	150	No
2	206.163.37.99	11:18:10	160.94.179.235	208	No
3	206.163.37.55	11:34:35	160.94.179.221	195	Yes
4	206.163.37.37	11:41:37	160.94.179.253	199	No
5	206.163.37.41	11:55:19	160.94.179.244	181	Yes



Anomaly Detection



Feature Extraction

- Three groups of features
 - **Basic features of individual TCP connections**
 - source & destination IP
 - source & destination port
 - Protocol
 - Duration
 - Bytes per packets
 - number of bytes
 - **Time based features**
 - For the same source (**destination**) IP address, number of unique destination (**source**) IP addresses inside the network *in last T seconds* –
 - Number of connections from source (**destination**) IP to the same destination (**source**) port *in last T seconds*
 - **Connection based features**
 - For the same source (**destination**) IP address, number of unique destination (**source**) IP addresses inside the network *in last N connections*
 - Number of connections from source (**destination**) IP to the same destination (**source**) port *in last N connections*

Fraud Detection

- Fraud detection refers to detection of criminal activities occurring in commercial organizations
 - Malicious users might be the actual customers of the organization or might be posing as a customer (also known as identity theft).
- Types of fraud
 - Credit card fraud
 - Insurance claim fraud
 - Mobile / cell phone fraud
 - Insider trading
- Challenges
 - Fast and accurate real-time detection
 - Misclassification cost is very high



Healthcare Informatics

- Detect anomalous patient records
 - Indicate disease outbreaks, instrumentation errors, etc.
- Key Challenges
 - Only normal labels available
 - Misclassification cost is very high
 - Data can be complex: spatio-temporal



Industrial Damage Detection

- Industrial damage detection refers to detection of different faults and failures in complex industrial systems, structural damages, intrusions in electronic security systems, abnormal energy consumption, etc.
 - Example: Aircraft Safety
 - Anomalous Aircraft (Engine) / Fleet Usage
 - Anomalies in engine combustion data
 - Total aircraft health and usage management
- Key Challenges
 - Data is extremely huge, noisy and unlabelled
 - Most of applications exhibit temporal behavior
 - Detecting anomalous events typically require immediate intervention



Computer Vision

- Detecting outliers in a image or video monitored over time
- Detecting anomalous regions within an image
- Used in
 - mammography image analysis
 - video surveillance
 - satellite image analysis
- Key Challenges
 - Detecting collective anomalies
 - Data sets are very large



Conclusions

- Anomaly detection can detect critical information in data.
- Highly applicable in various application domains.
- Nature of anomaly detection problem is dependent on the application domain.
- Need different approaches to solve a particular problem formulation.